Lab 5

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1. Code Analysis
    - ga.py implements a genetic algorithm
    - Each individual is a genome of 50 real valued weights (24 sensors x 2 outputs + 2 biases)
    - The genome is stored in the environment variable GENOME
    - ARGoS is launched through evaluate-controller-nn.argos
    - Controller-nn.lua reads the genome and builds a neural network using nn.lua
    - Fitness is computed from the robot's distance to the light source and printed
    - ga.py captures the fitness and uses it for evolution
2. Random Solutions Testing
    - Random genomes were generated and evaluated, robots showed random and inconsistent behavior. This improved when implemented mutation operator
3. Qualitative Evaluation
    - The algorithm was successful in improving fitness over generations
    - Fitness values above 0.9 were frequently achieved after a few generations
    - Example runs reached fitness values between 0.97 - 0.98 which indicates success in evolutionary process to learn controllers that move toward the light source
4. Role of N_EVAL
    - N_EVAL determines how many times each robot (controller) is evaluated
    - When multiple evaluations are present (like 3 in the code) it reduces the effect of randomness in the simulation
    - When changed to 1 the execution was faster but fitness becomes more biased and when N_EVAL=3 fitness estimation is more reliable
5. Mean vs Minimum Fitness
    - Mean: rewards average performance and produces higher fitness values
    - Min: penalizes unreliable controller and produces lower fitness values
    - Good solutions can still be found in Minimum
6. Crossover Rate Comparison
    - All crossover rates achieved high fitness values
    - Results were stochastic and contained occasional poor runs
    - Median values were more stable than mean because they are less affected by outliers
    - Unclear which Crossover Rate was clearly superior using only 5 replicas
7. Replacement vs Elitism
    - Replacement performed better than elitism in experiments

    - Elitism guarantees preservation of genomes, but not preservation of exact fitness value as the controller is re-evaluated in each run
    - Elitism sometimes would preserve mediocre individuals
8. Tournament vs Roulette Wheel Selection
    - Roulette produced better results
    - Tournament selection applies more pressure in terms of selection
    - Roulette wheel selection maintained more diversity and produced more high quality solutions from conducted tests